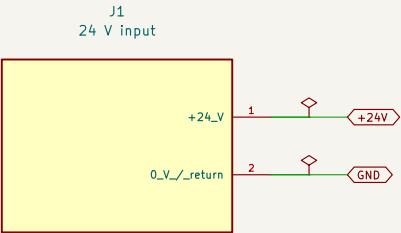
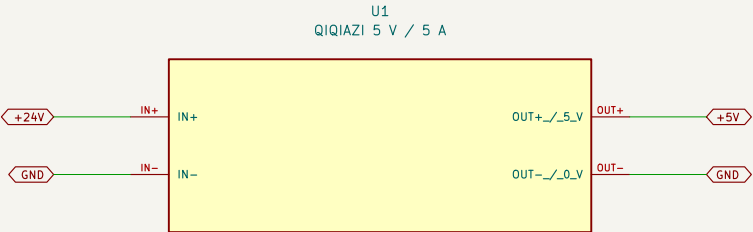


CONTROLLER AND POWER

A



B



Wire converter by IN+ / IN- / OUT+ / OUT- markings.
24 V also feeds all three motor drivers.
Common 0 V; separate power branches to each driver.
5 A is the converter rating, not the Nucleo input rating.

C

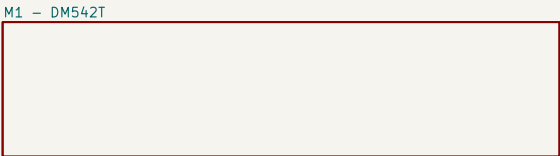
NUCLEO POWER CONFIGURATION
JP2: fit EXT jumper on pins 5–6.
Buck +5 V -> CN11 pin 6; ground -> CN11 pin 8.
CN8 pin 7 supplies 3.3 V to interface/storage.
5V_EXT input: 4.75–5.25 V, 500 mA maximum.
Power EXT first, then connect ST–Link USB CN1.
Do not connect 24 V to Nucleo VIN or logic pins.
STLINK–V3 MCO: HSE/5 = 5 MHz (persistent setting).
Target HSE bypass; CPU 480 MHz / timer kernels 240 MHz.

D

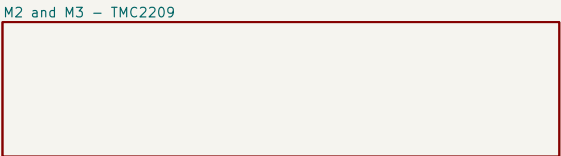
BOARD ROUTING / TIM2 DMA STEP OUTPUTS
PA0 -> CN10.29; SB75 ON; user button stays PC13.
QSPI: SB59/SB63 ON; SB61/SB69 OFF.
PE2 -> CN10.25; SB67 OFF.
Only used external contacts are drawn.
Ethernet, ST–Link, LEDs and button stay onboard.

Shared names connect across sheets. A1 contacts use CNxx.pin numbering. J101–J111 are perfboard headers.
REF inputs: internal 3.3 V pull-ups, both-edge EXTI; NC to GND. Released = LOW; pressed / unplugged = HIGH.
U1/A3/A4 mount at J103/J105/J106. Buck: 24 V via barrel jack; VIN pins support only. TMC power/motor cables on JP1.

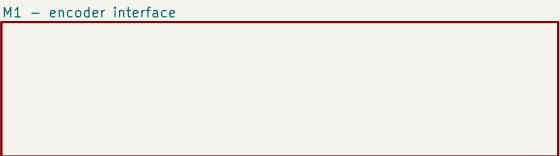
E



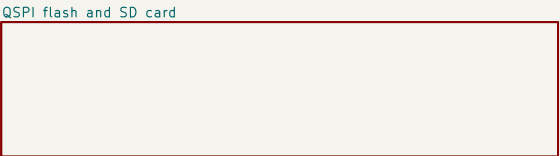
Date: sheets/M1_DM542T.kicad_sch



Date: sheets/M2_M3_TMC2209.kicad_sch

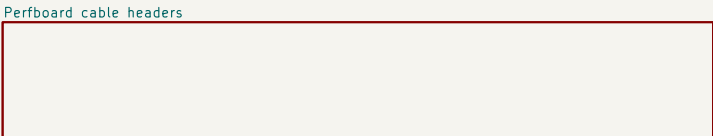


Date: sheets/M1_Encoder.kicad_sch

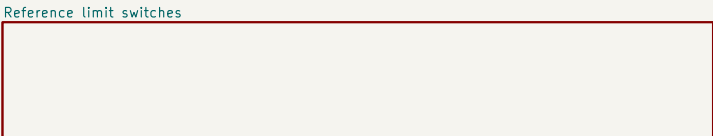


Date: sheets/Storage.kicad_sch

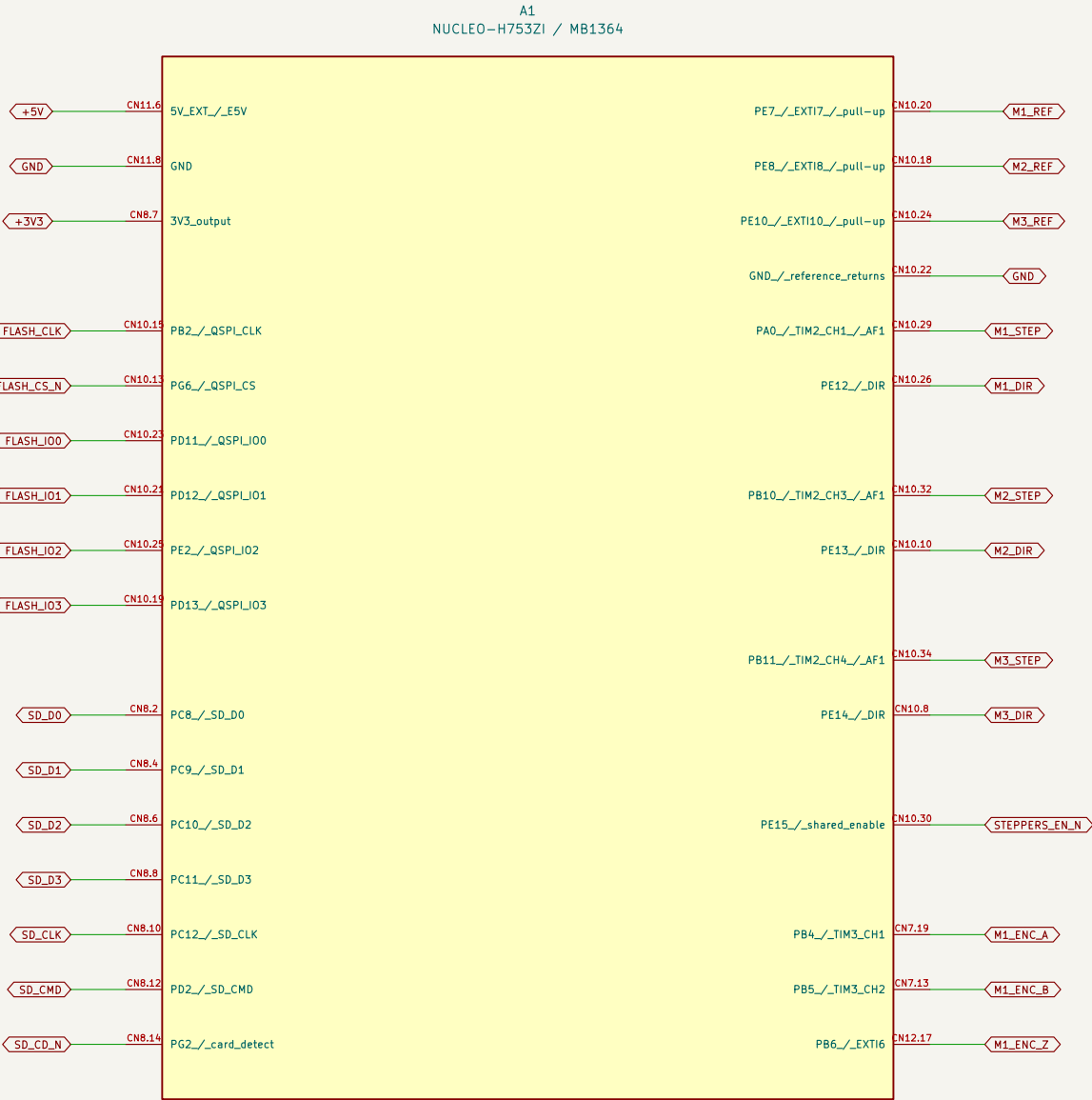
F



Date: sheets/Perfboard_Connectors.kicad_sch



Date: sheets/Reference_Switches.kicad_sch



System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

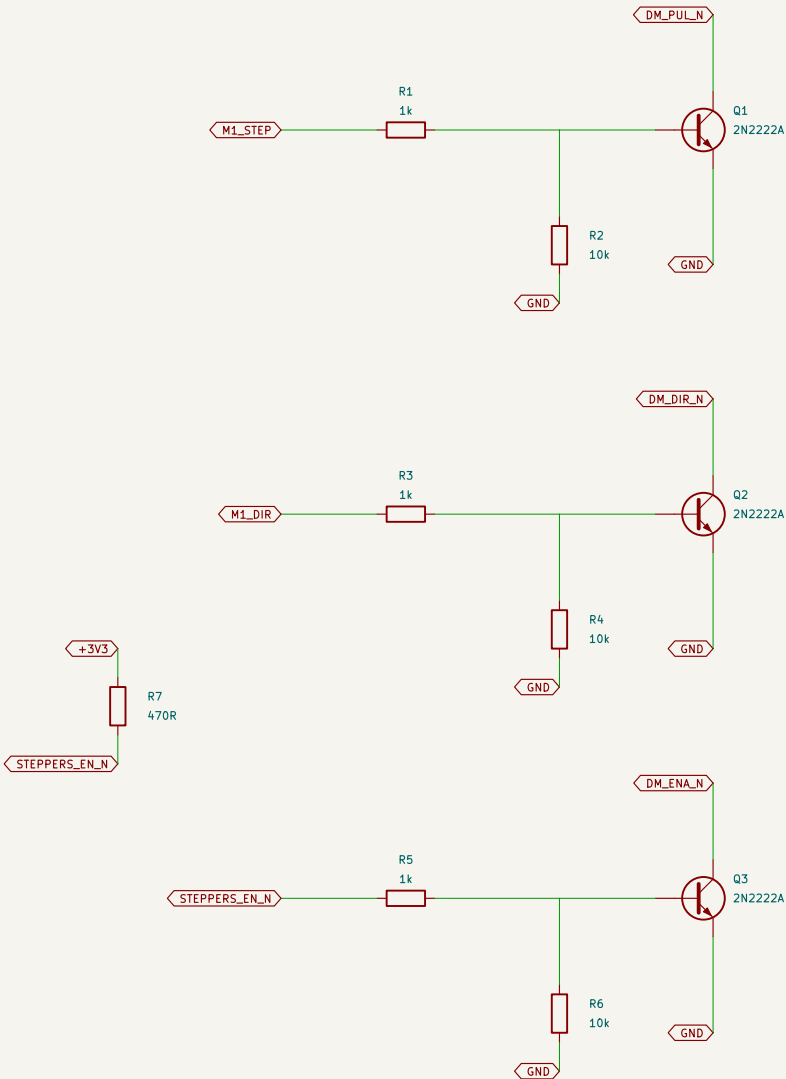
Sheet: /
File: ProtoMate.kicad_sch

Title: ProtoMate SCARA – Controller and power

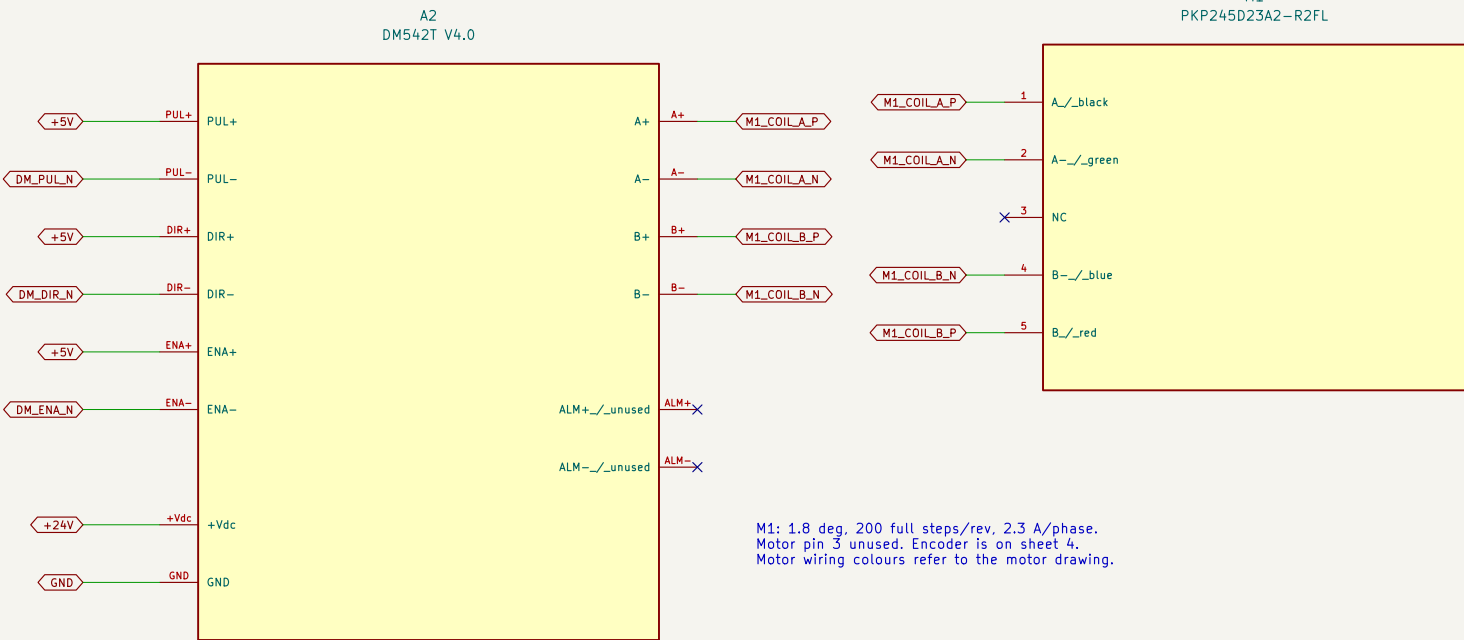
Size: A3	Date: 2026–10–03	Rev: 5.0
KiCad E.D.A. 10.0.6		Id: 1/7

M1 – DM542T

3.3 V MCU -> 2N2222A current sinks -> 5 V optoisolated inputs



SHARED ENABLE AND TIMING
PE15 low: Q3 off, DM542T enabled, both TMC EN inputs low.
PE15 high: Q3 on, DM542T disabled, both TMC EN inputs high.
R7 pulls PE15 high during MCU reset; about 7 mA sink when enabled.
After enable, wait >=200 ms. DIR setup >=5 us; STEP high/low >=2.5 us.



DM542T SETTINGS / INITIAL COMMISSIONING
S2 signal selector: 5 V.
1/16 step: SW5 ON, SW6 ON, SW7 OFF, SW8 ON.
3200 STEP pulses per motor revolution.
Conservative initial current: 2.03 A RMS / 2.84 A peak
SW1 ON, SW2 ON, SW3 OFF; SW4 OFF (idle reduction).
Check the actual V4.0 label before setting switches.

Q1–Q3 show functional B/C/E pins, not package lead numbers.
Verify your 2N2222A manufacturer/package before assembly.
Fit each 10k between base and emitter. Keep PUL+/PUL- paired;
separate signal cables from motor power cables.

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control		
Sheet: /M1 – DM542T/ File: M1_DM542T.kicad_sch		
Title: ProtoMate SCARA – M1 – DM542T		
Size: A3	Date: 2026–10–02	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 2/7

M2 AND M3 – TMC2209

MS1 = MS2 = VDD (3.3 V): 1/16 step, 3200 pulses/rev. No MCU pins needed.
SPRD jumper open: factory StealthChop mode. Current set with each onboard potentiometer.

A

B

C

D

E

F

A

B

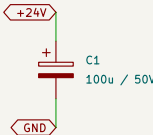
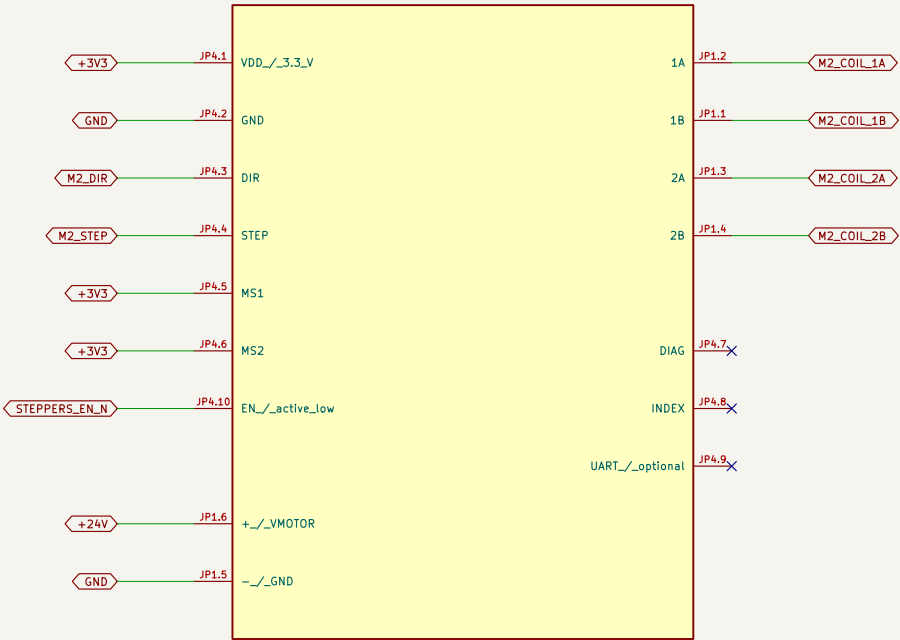
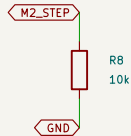
C

D

E

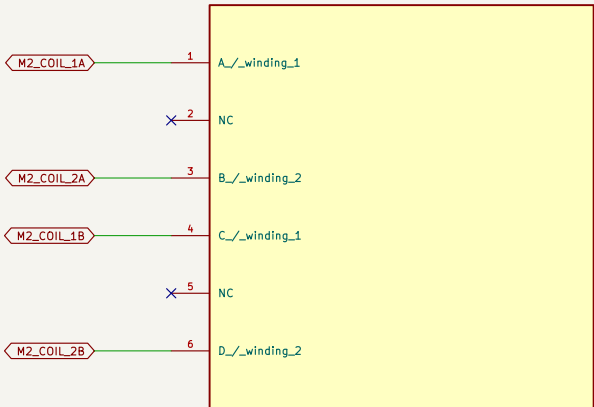
F

A3
Adafruit #6121 / TMC2209

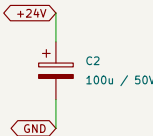
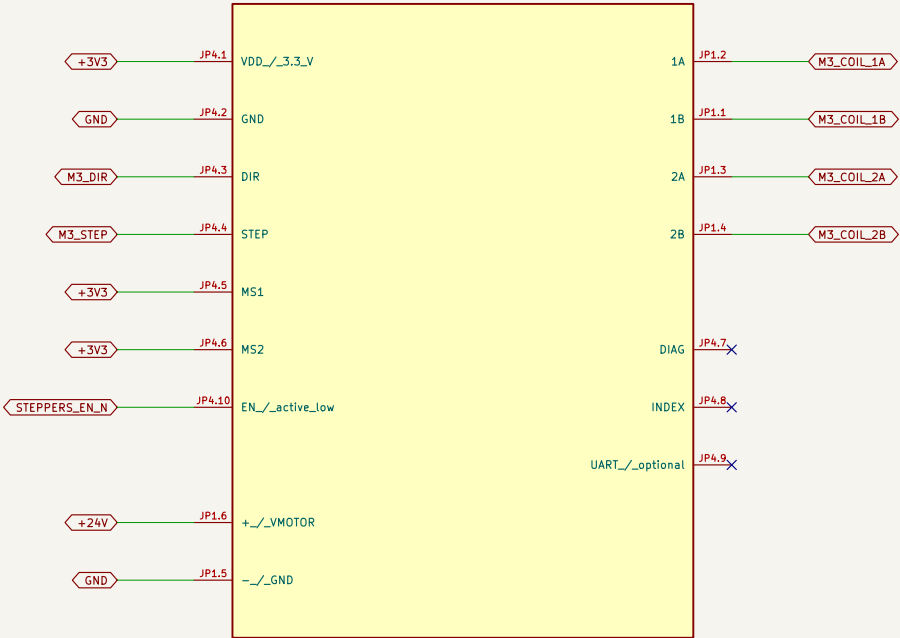
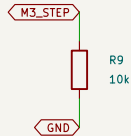


Extra bulk capacitor near VMOTOR.
Onboard bypass/bulk already included.
Keep VMOTOR below 29 V including
deceleration transients.

M2
C17HD2024-01N

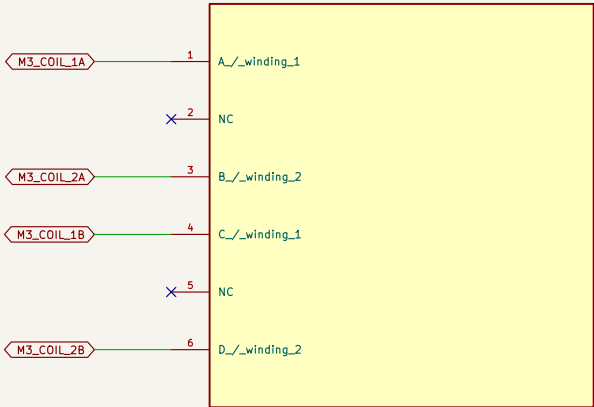


A4
Adafruit #6121 / TMC2209



Extra bulk capacitor near VMOTOR.
Onboard bypass/bulk already included.
Keep VMOTOR below 29 V including
deceleration transients.

M3
BJ42D15-26V10



M2: 1.0 A/phase, 4.1 ohm, 8.5 mH; coil pairs 1-4 and 3-6 (user-confirmed).
M3: 0.84 A/phase, 6.0 ohm, 8.8 mH: 20T / 2 mm pulley -> 40 mm/rev -> 80 pulses/mm.
Both motors: 1.8 deg. Adjust current separately: respect peak/RMS conventions.
A3/A4 mount at J105/J106 via JP4 headers: JP1 power/motor cables stay directly on each module.

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /M2 and M3 – TMC2209/

File: M2_M3_TMC2209.kicad_sch

Title: ProtoMate SCARA – M2 and M3 – TMC2209

Size: A3

Date: 2026-10-02

Rev: 1.0

KiCad E.D.A. 10.0.6

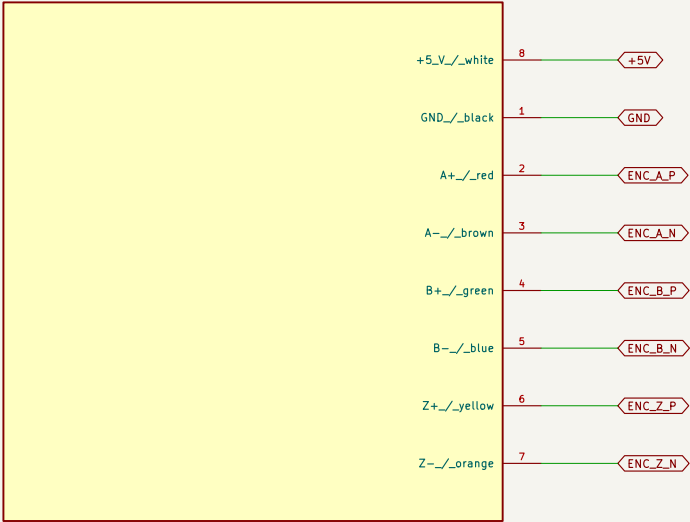
Id: 3/7

M1 – ENCODER / HC126 + NPN LEVEL SHIFTERS

U2 input pairs are deliberately reversed.
Q4–Q6 cancel that inversion before U3.
Encoder connector pinout stays unchanged.
Never wire U2 outputs directly to U3.

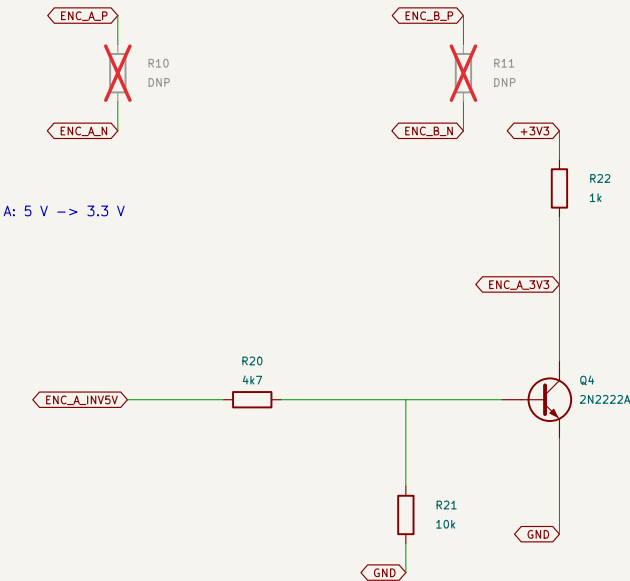
U2: 5 V. Q4–Q6 translate to 3.3 V. U3: 3.3 V ONLY.
U3 OE1/OE2/OE3 HIGH; unused OE4 and A4 LOW.
U2 unused receiver inputs may float (fail–safe).
100 nF at each IC; 1 uF nearby.

J2
M1 encoder / 8–pin



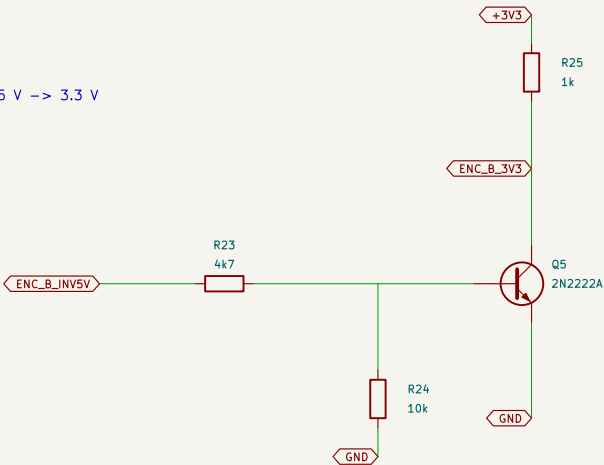
400 P/R -> TIM3 x4 = 1600 counts/rev; Z = once/rev.
Cable colours / connector orientation; motor manual.
Shielded twisted pairs; chassis bond at cable entry.

R10–R12: OPTIONAL TERMINATION, DNP.
Choose for cable and 20 mA encoder output limit.
A+/A-, B+/B-, Z+/Z-: twisted pairs.

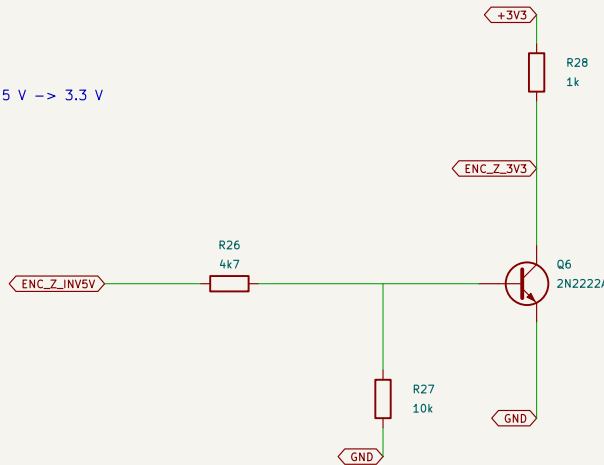


A: 5 V -> 3.3 V

B: 5 V -> 3.3 V



Z: 5 V -> 3.3 V



A/B -> PB4/PB5; Z -> PB6 rising–edge EXTI. All original signal polarities preserved.
Q4–Q6: Diotec 2N2222A E=1/B=2/C=3. Keep collector–to–U3 wires short.
Check edge shape and transistor turn–off delay at the intended encoder speed.

System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /M1 – encoder interface/
File: M1_Encoder.kicad_sch

Title: ProtoMate SCARA – M1 – encoder interface

Size: A3

Date: 2026–10–03

Rev: 2.0

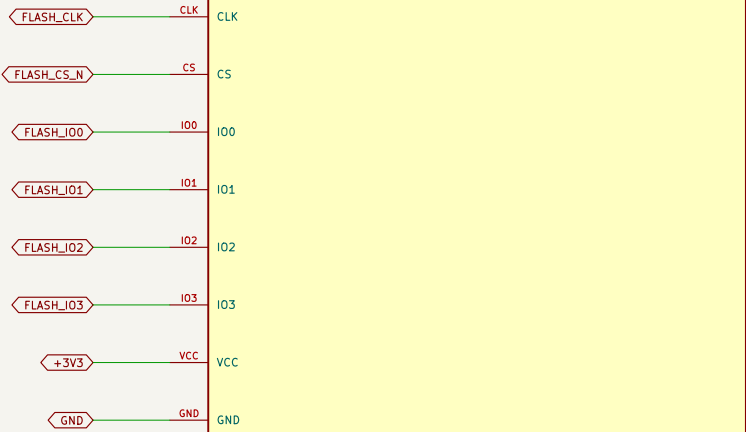
KiCad E.D.A. 10.0.6

Id: 4/7

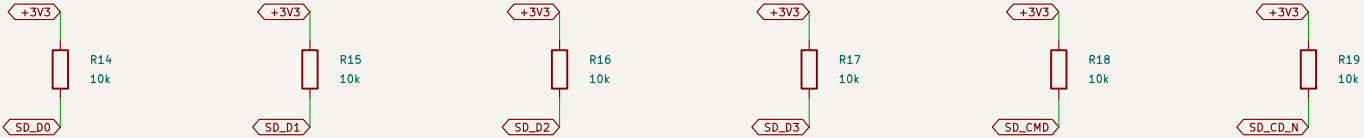
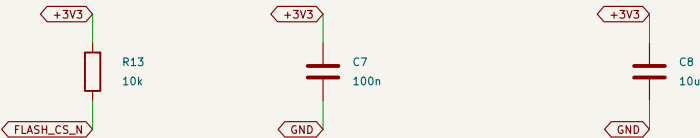
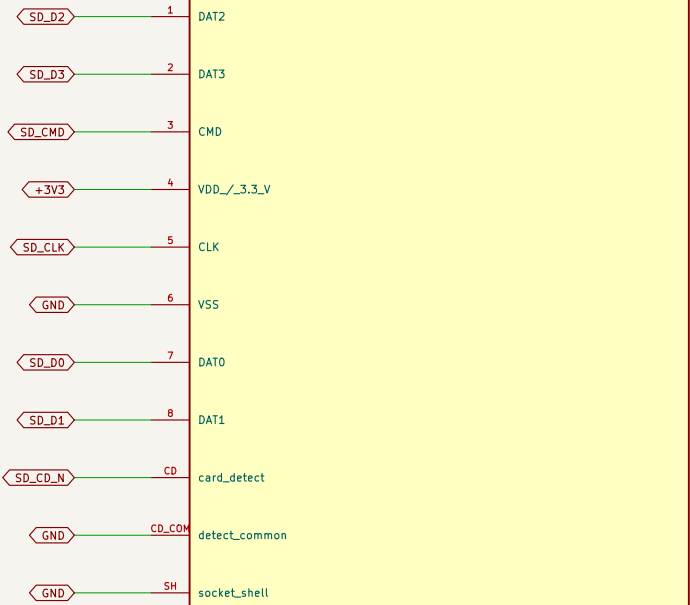
QSPI FLASH AND SD CARD

SD pins 1–8 use standard card–contact numbering.
CD switch closes to GND when inserted (advisory input).
Pull–ups on CMD/D0–D3/CD; no pull–up on CLK.
Existing equivalent module pull–ups/caps need not be duplicated.

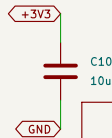
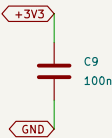
A5
W25Q128 / 3.3 V QSPI



J3
microSD / 4–bit socket module



3.3 V ONLY
Use a W25Q128 module exposing IO0–IO3.
IO2/WP and IO3/HOLD are data in quad mode;
do not hard–wire them to supply or ground.
Module contacts are identified by signal name.
C7/C8 and R13 may already exist on the module;
fit them if absent. Keep the QSPI harness short.



System wiring and linked perfboard assembly map | all voltages DC

ProtoMate Control

Sheet: /QSPI flash and SD card/

File: Storage.kicad_sch

Title: ProtoMate SCARA – QSPI flash and SD card

Size: A3

Date: 2026–10–02

Rev: 1.0

KiCad E.D.A. 10.0.6

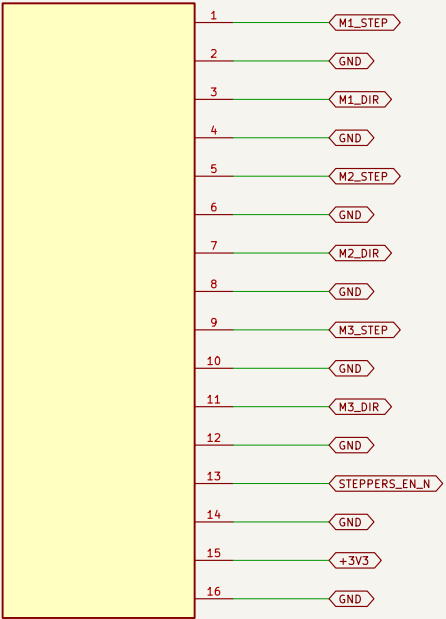
Id: 5/7

PERFBOARD HEADERS / MODULE MOUNTS

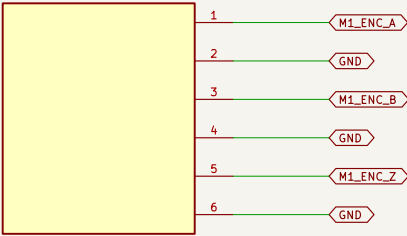
Pin numbers match the layout and Harness.csv. Contacts use the 2.54 mm grid; J103 pairs span 5.08 mm. Mount pins and cable pins are shown.

A

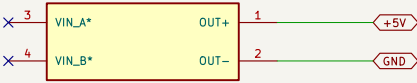
J101
Nucleo outputs / 2x08



J102
Nucleo encoder / 2x03

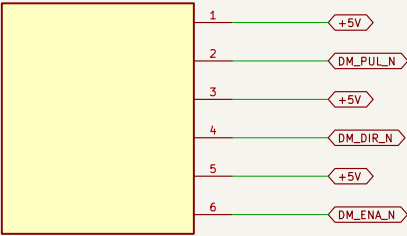


J103
U1 buck direct mount



* Pads 3/4: VIN support pins, live via U1; NO perfboard wires.
24 V enters U1 through its barrel jack.
Pins 1/2: regulated OUT+ / OUT-.

J104
DM542T / 1x06



J101/J102: paired even pins are GND.
J101 pin 15 receives Nucleo 3.3 V.
J103: U1 mount; OUT+/OUT- feed 5 V/GND.
24 V / motor wiring goes directly to the modules.

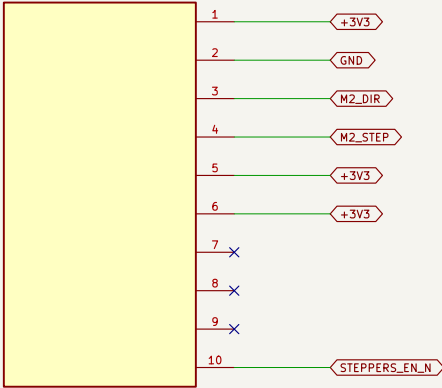
J104: 1 PUL+, 2 PUL-, 3 DIR+, 4 DIR-, 5 ENA+, 6 ENA-.
J105/J106: 1 VDD, 2 GND, 3 DIR, 4 STEP, 5 MS1, 6 MS2,
7 DIAG (NC), 8 INDEX (NC), 9 UART (NC), 10 EN.
J107 follows the numbered J2 encoder connector.

B

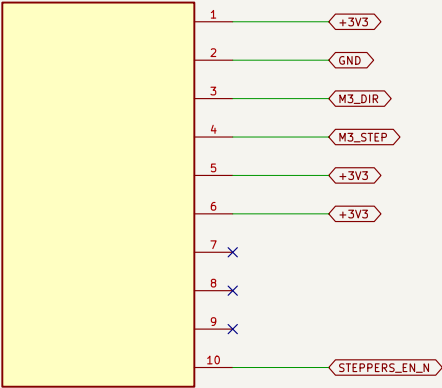
C

D

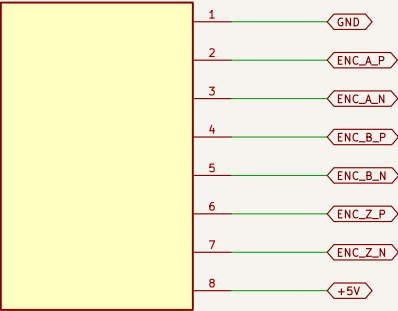
J105
M2 TMC2209 direct mount



J106
M3 TMC2209 direct mount



J107
M1 encoder / 1x08



J101 -> Nucleo CN10: 1->29, 3->26, 5->32, 7->10, 9->34, 11->8, 13->30.
J103/J105/J106 are direct module mounts, no cable. Other headers use cables. Matching names are one conductor.
J108-J111 (Nucleo reference inputs and the three NC switches): see the Reference limit switches sheet.

All J101-J107 headers mount on the perfboard; signals connect by global names

ProtoMate Control

Sheet: /Perfboard cable headers/

File: Perfboard_Connectors.kicad_sch

Title: ProtoMate SCARA - Perfboard cable headers

Size: A3

Date: 2026-10-03

Rev: 3.0

KiCad E.D.A. 10.0.6

Id: 6/7

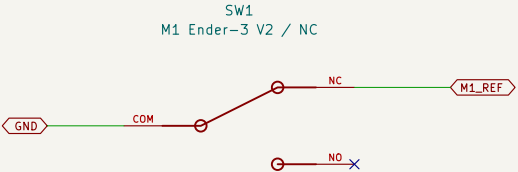
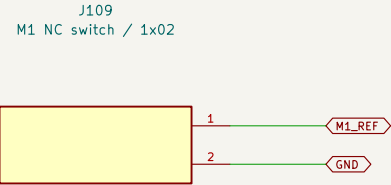
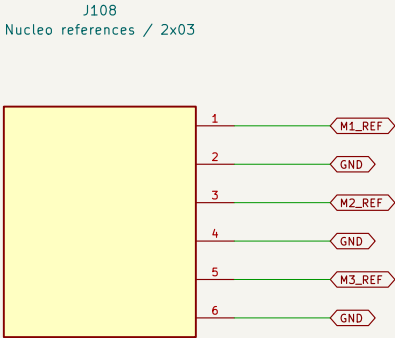
F

F

REFERENCE LIMIT SWITCHES – NORMALLY CLOSED

J108–J111 mount on the perfboard. SW1–SW3 mount on the robot. Global labels connect the cable endpoints.

Switches shown released (COM–NC closed).



J108 -> Nucleo A1
1 = PE7 / CN10.20 (M1)
3 = PE8 / CN10.18 (M2)
5 = PE10 / CN10.24 (M3)
2, 4, 6 = GND / CN10.22

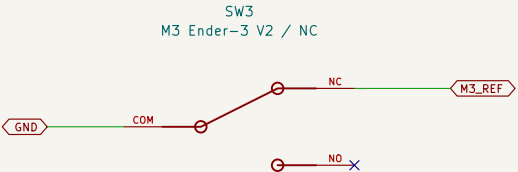
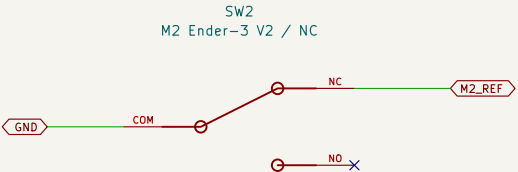
INPUT BEHAVIOR
Nucleo internal pull–ups to 3.3 V.
Released: COM–NC closed -> LOW.
Pressed / unplugged: open -> HIGH.
Both–edge EXTI: raw, un–debounced inputs.
Homing and motor–stop logic not yet implemented.

BEFORE WIRING: With the switch disconnected, verify continuity: closed when released, open when pressed.
COM / NC / NO identify contact functions, NOT the module connector pin order. Do not infer power wiring from V/G/S printing.
Use the unpopulated mechanical version as a dry contact; no supply connection. Route each REF with its GND return, away from motor wiring.

J109.1 -> verified NC contact
J109.2 -> verified COM contact
NO contact stays unconnected.

J110.1 -> verified NC contact
J110.2 -> verified COM contact
NO contact stays unconnected.

J111.1 -> verified NC contact
J111.2 -> verified COM contact
NO contact stays unconnected.



NC dry contacts | PE7 / PE8 / PE10 | 3.3 V internal pull–ups

ProtoMate Control

Sheet: /Reference limit switches/
File: Reference_Switches.kicad_sch

Title: ProtoMate SCARA – Reference limit switches

Size: A3 Date: 2026–10–03 Rev: 4.0
KiCad E.D.A. 10.0.6 Id: 7/7